

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 9: Orbital Calibration

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Overview

Orbital residuals are expressed in terms of apsidal and nodal precession functions before fitting the relevant parameter block.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\Delta\varpi = \beta \cdot 6\pi GM/[a(1 - e^2)c^2]$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$\beta = 1.84$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$\Delta\Omega = \alpha_3 f_3(a, e, i) + \alpha_4 f_4(a, e, i)$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper C, §§6–7
- SGT9 orbital block P-01–P-14
- Ephemeris Standard ES-29